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**5.** The rock pocket mouse lives in a sandy desert in the United States. The mouse may be either light or dark in colour. A volcano erupted 1.7 million years ago and turned large parts of the desert into dark coloured rock. **Image 5.1** shows how two locations (**A** and **B**) in the desert could have looked before the volcano erupted.

**Image 5.1**

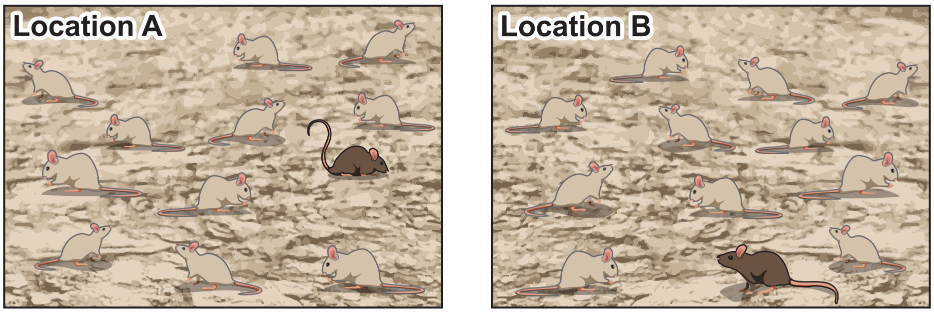
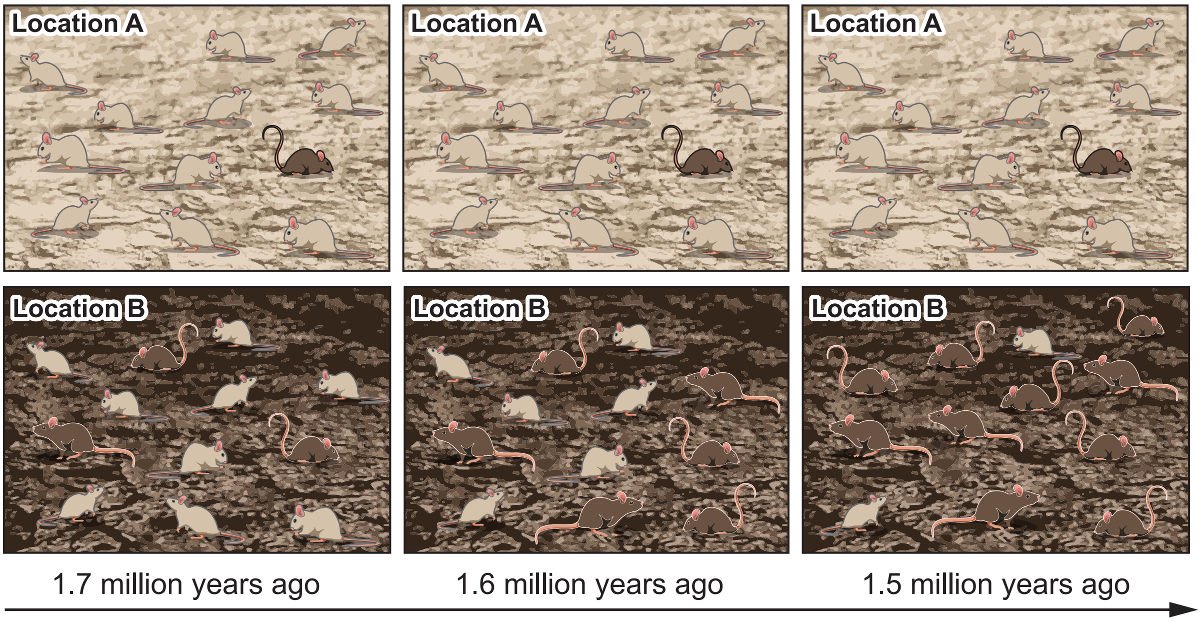



**Image 5.2** shows how the mouse population might have changed over time after the volcano erupted.

**Image 5.2**




(a) Circle the cause of the changed rock colour in location **B** from the list below. [1]

**flooding** **volcanic eruptions** **human activities** **forest fires**



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(b) **Tables 5.3** and **5.4** show the number of light and dark coloured mice over time.

**Table 5.3** – Location **A**

| | Number of mice in Location **A** | | |
|---|---|---|---|
| Time period (million years ago) | light coloured | dark coloured | Total |
| 1.7 | 11 | 1 | 12 |
| 1.6 | 11 | 1 | 12 |
| 1.5 | 11 | 1 | 12 |

**Table 5.4** – Location **B**

| | Number of mice in Location **B** | | |
|---|---|---|---|
| Time period (million years ago) | light coloured | dark coloured | Total |
| 1.7 | 9 | 3 | .................... |
| 1.6 | .................... | 6 | 12 |
| 1.5 | 2 | .................... | 12 |

(i) Use **Image 5.2** to complete **Table 5.4**. [2]



 

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(ii) Use the information in **Table 5.4** to complete **Graph 5.6** for Location **B** by

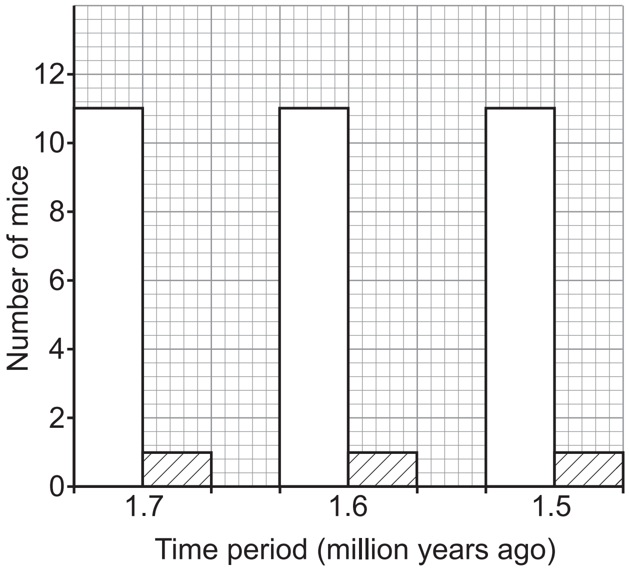
I. adding the axes labels; [1]

II. adding a scale to both axes; [1]

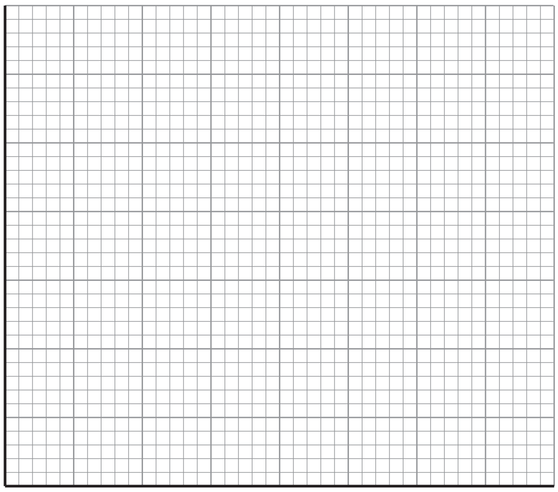
III. drawing bars for each of the results. [2]

Location **A** has been done for you on **Graph 5.5**.

**Graph 5.5** – Location **A**




**Graph 5.6** – Location **B**



**Key:** light coloured mice | dark coloured mice

(iii) State what happened over time to the number of **light coloured** mice at

I location **A**; [1]

..........................................................................................................................................

II location **B**. [1]

..........................................................................................................................................



 (3400U20-1) **Turn over.**

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(c) (i) A mutation caused the dark-coloured mice to appear in a population of light coloured mice.
Define the term **mutation**. [1]

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(ii) Suggest and explain why the dark coloured mice had an advantage in Location **B**. [2]

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(iii) Suggest what will happen to the number of light coloured mice in Location **B** in the future. [1]

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(iv) The change in colour over time is an example of natural selection or survival of the fittest.
Tick (✓) the boxes next to the **two** scientists who developed the theory of natural selection. [2]

| | |
|---|---|
| Gregor Mendel | ☐ |
| Charles Darwin | ☐ |
| Carl Linnaeus | ☐ |
| Alfred Wallace | ☐ |
| Alexander Fleming | ☐ |

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